

Ptychographic transmission microscopy in three dimensions using a multi-slice approach

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Generally, methods of three-dimensional imaging such as confocal microscopy and computed tomography rely on two essentials: multiple measurements (at a range of focus positions or rotations) and a weakly scattering specimen (to avoid distortion of the focal spot in the confocal microscope or to satisfy the projection approximation in tomography). Here we show that an alternative form of multi-measurement imaging, ptychography, can be extended to three dimensions and can successfully recover images in the presence of multiple scattering and when the projection approximation is not applicable. We demonstrate our technique experimentally using visible light, where it has applications in imaging thick samples such as biological tissues; however the results also have important implications for x ray and electron imaging. © 2012 Optical Society of America

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1. INTRODUCTION

Conventional two-dimensional (2D) ptychography entails a specimen (the object), described by a complex transmission function $O(\mathbf{r})$, illuminated by a patch of radiation (the probe) described by the 2D stationary complex function $P(\mathbf{r})$, where $\mathbf{r} = (x, y)$ is a Cartesian coordinate. The probe, or equivalently the specimen, is shifted to an array of positions $\mathbf{R}_j = (R_{x,j}, R_{y,j})$, and $j = 1 \dots J$ diffraction patterns are recorded some distance downstream. (Because the probe in ptychography is many times larger than the resolution of the reconstructed image, J is typically in the middle hundreds.) Hoppe realized [1] that if the shifts are arranged such that the area of specimen covered by adjacent probes overlap, the redundancy thereby introduced into the recorded data allows solution of the phase problem. From this beginning, ptychography has developed into a phase retrieval technique particularly popular among the x ray community [2–6], and has recently been demonstrated using electrons [7–9]. One of the keys to this success is that iterative ptychographic algorithms can now robustly reconstruct images of both $P(\mathbf{r})$ and $O(\mathbf{r})$ from a single data set [10–12], thereby minimizing the required *a priori* knowledge of the experimental setup.

Unlike previous work combining phase retrieval with specimen rotations [13–15], we report here a three-dimensional (3D) extension to ptychography that uses identical data to that of the conventional 2D technique—indeed our method can be applied retrospectively to existing data. Alternative methods using solely ptychographic data have yielded only a projection of the specimen from several angles [16,17], or have been limited to crystalline materials where an x-ray Bragg peak is processed [18].

Crucially our method takes account of 3D- and multiple-scattering effects when modelling the propagation of the probe through the specimen. Even in the absence of multiple scattering, conventional 2D ptychography cannot handle a

thick 3D specimen whose interaction with the probe cannot be modeled using a multiplicative approximation. By causing a breakdown of the projection approximation, 3D effects similarly limit specimen thickness in x ray and electron computed tomography [14,19], and multiple scattering also limits the depth within a specimen that confocal microscopy can reach [20,21]. In digital holography an exit wave is reconstructed that can be propagated back and forth, but again multiple scattering prevents optical sectioning of this wavefront [22] (although some advances have been reported that improve upon this situation, e.g., [23]). By accounting for 3D and multiple scattering processes, our method can be applied to thick specimens that cannot be imaged using these existing techniques.

2. RECONSTRUCTION ALGORITHM

A. Outline

Our algorithm, which we call 3PIE [24], improves upon the conventional multiplicative approximation used by 2D ptychography. Instead we use the multi-slice approach widely used in electron microscopy [25,26], which models the specimen as a number of slices or layers. As for 2D ptychography, the exit wave from the first (most upstream) layer is calculated as the product of the incident probe and the complex-valued transmission function for that layer. However instead of being propagated to the detector plane, this wavefront is then propagated to, and becomes the incident probe for, a second layer. The multiplicative approximation is used again to estimate the wave function exiting this second layer, which is propagated to the next layer, and so on and so forth. In the inverse part of the calculation (after the usual diffraction intensity constraint has been applied), the propagation steps are reversed, with each layer of the object and the illumination functions incident upon each layer updated in a similar

way to the extended Ptychographical Iterative Engine (ePIE) [12].

B. Step-by-Step Description

The multiplicative approximation used for 2D ptychography assumes that the wavefront, $\Psi_j(\mathbf{u})$, whose intensity is measured as the j th recorded diffraction pattern is

$$\Psi_j(\mathbf{u}) = \mathcal{F}[P(\mathbf{r} - \mathbf{R}_j) \cdot O(\mathbf{r})], \quad (1)$$

where $\mathcal{F}$ is the Fourier propagator and $\mathbf{u}$ is a coordinate in the diffraction plane.

Equation (1) can only accommodate specimens that are sufficiently thin [11]—to overcome this limitation the 3PIE models a thick specimen using $n = 1 \dots N$ slices with each slice representing a volume within the specimen for which the multiplicative approximation is valid [as shown in Fig. 1(a) for the case $N = 3$]. Labelling the n th slice, which is located at $z = z_n$, by $O_n(\mathbf{r})$, the improved model of $\Psi_j(\mathbf{u})$ is

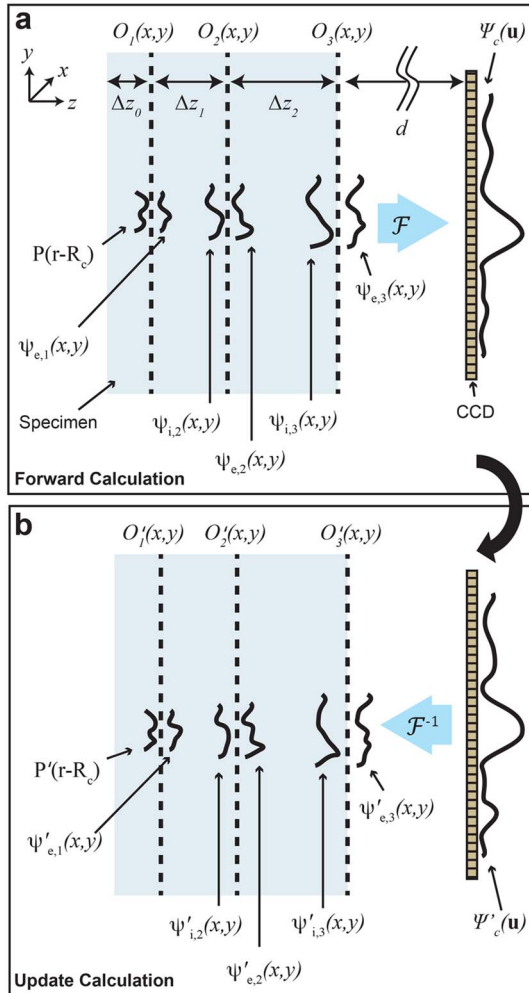


Fig. 1. (Color online) Schematic detailing the nomenclature used to describe the 3PIE method. (a) The algorithm performs a multi-slice calculation to arrive at an approximation to the wavefront incident upon the detector; (b) This wavefront is modified to match the recorded data and the result used to update each of the slices used in the forward multi-slice calculation.

$$\Psi_j(\mathbf{u}) = \mathcal{F}[O_3(\mathbf{r}) \cdot \mathcal{P}_{\Delta z_2}[O_2(\mathbf{r}) \cdot \mathcal{P}_{\Delta z_1}[P(\mathbf{r} - \mathbf{R}_j) \cdot O_1(\mathbf{r})]]], \quad (2)$$

for the case $N = 3$. Here the propagator $\mathcal{P}_{\Delta z_n}$ is an angular spectrum propagator over the distance, $\Delta z_n = z_{n+1} - z_n$, between the n th and $(n + 1)$ th slices. The extension for $N > 3$ continues this multiply-propagate pattern.

The inputs to the 3PIE are the J recorded diffraction patterns together with an initial estimate of the probe and initial estimates of each slice of the object. These need be nothing more than a circle of uniform intensity and roughly the correct size for the probe, and free-space for each object slice. The algorithm is described by the following steps:

1. Randomly select one of the recorded diffraction patterns that has not been addressed in the current iteration. We will label this pattern $I_c(\mathbf{u})$, to denote it as the current pattern, and label the corresponding offset of the probe $\mathbf{R}_c = (R_{x,c}, R_{y,c})$.

Referring to Fig. 1(a), an estimate of the wavefront that resulted in $I_c(\mathbf{u})$ is obtained as follows:

2. Compute the exit wave from the first slice as $\psi_{e,1}(\mathbf{r}) = P(\mathbf{r} - \mathbf{R}_c) \cdot O_1(\mathbf{r})$.

3. Propagate this wave to the second slice, creating a subsequent incident wave $\psi_{i,2}(\mathbf{r}) = \mathcal{P}_{\Delta z_1}[\psi_{e,1}(\mathbf{r})]$, where $\mathcal{P}_{\Delta z_n}$ is an angular spectrum propagator over distance $\Delta z_n = z_{n+1} - z_n$.

4. Compute the exit wave from the second slice as $\psi_{e,2}(\mathbf{r}) = \psi_{i,2}(\mathbf{r}) \cdot O_2(\mathbf{r})$, and propagate this wavefront to the third slice to give $\psi_{i,3}(\mathbf{r}) = \mathcal{P}_{\Delta z_2}[\psi_{e,2}(\mathbf{r})]$.

5. Continue this multiply-propagate process through subsequent slices until the exit-wave for the final slice, $\psi_{e,N}(\mathbf{r})$, has been reached.

6. Propagate this wavefront to the detector to give $\Psi_c(\mathbf{u}) = \mathcal{F}[\psi_{e,N}(\mathbf{r})]$, where the propagator in this instance is a Fourier or Fresnel propagator over the distance d .

Care is required in step 6 because unlike 2D ptychography a curvature cannot appear across the reconstructed probe to compensate for a detector closer to the specimen than the far-field conditions allow; therefore the correct propagator must be selected.

Referring now to Fig. 1(b), steps 7 and 8 generate a revised estimate of the wavefront exiting the specimen, $\psi'_{e,N}$, exactly as for conventional ptychography:

7. Correct the modulus of $\Psi_c(\mathbf{u})$ according to the measured diffraction pattern $I_c(\mathbf{u})$ to give

$$\Psi'_c(\mathbf{u}) = \sqrt{I_c(\mathbf{u})} \frac{\Psi_c(\mathbf{u})}{|\Psi_c(\mathbf{u})|}.$$

8. Back-propagate the corrected wavefront to the plane of the N th object slice, giving a revised estimate of $\psi'_{e,N}$:

$$\psi'_{e,N} = \mathcal{F}^{-1}[\Psi'_c(\mathbf{u})].$$

The forward multi-slice calculation must now be unravelled in order to update each slice of the current object estimate and the probe. This is accomplished using an update function that can be posed in terms of three functions and a single constant parameter α as

$$\mathcal{U}[f(\mathbf{r}), g(\mathbf{r}), \Delta\psi(\mathbf{r})] = f(\mathbf{r}) + \alpha \frac{g^*(\mathbf{r})}{|g(\mathbf{r})|_{\max}^2} \Delta\psi(\mathbf{r}). \quad (3)$$

This is an amalgamation of the two update functions used by the ePIE. Using Eq. (3), the update computation proceeds by the following “update-back propagate” steps:

9. Set $\Delta\psi(\mathbf{r}) = \psi'_{e,N}(\mathbf{r}) - \psi_{e,N}(\mathbf{r})$ and update the N th incident wavefront and the N th object slice according to

$$\begin{aligned} \psi'_{i,N}(\mathbf{r}) &= \mathcal{U}[\psi_{i,N}(\mathbf{r}), O_N(\mathbf{r}), \Delta\psi(\mathbf{r})], \\ O'_N(\mathbf{r}) &= \mathcal{U}[O_N(\mathbf{r}), \psi_{i,N}(\mathbf{r}), \Delta\psi(\mathbf{r})]. \end{aligned}$$

10. Back propagate the revised incident wave, $\psi'_{i,N}(\mathbf{r})$, to the plane of the $(N-1)$ th object slice:

$$\psi'_{e,N-1}(\mathbf{r}) = \mathcal{P}_{-\Delta z_{(N-1)}}[\psi'_{i,N}(\mathbf{r})].$$

11. Set $\Delta\psi(\mathbf{r}) = \psi'_{e,N-1}(\mathbf{r}) - \psi_{e,N-1}(\mathbf{r})$ and update the $(N-1)$ th incident wave and object slice:

$$\begin{aligned} \psi'_{i,N-1}(\mathbf{r}) &= \mathcal{U}[\psi_{i,N-1}(\mathbf{r}), O_{N-1}(\mathbf{r}), \Delta\psi(\mathbf{r})], \\ O'_{N-1}(\mathbf{r}) &= \mathcal{U}[O_{N-1}(\mathbf{r}), \psi_{i,N-1}(\mathbf{r}), \Delta\psi(\mathbf{r})]. \end{aligned}$$

12. Repeat steps 10 and 11 for slices $N-2, N-3, \dots$ until the first slice is reached.

13. Set $\Delta\psi(\mathbf{r}) = \psi'_{e,1}(\mathbf{r}) - \psi_{e,1}(\mathbf{r})$ and update the probe and the first slice of the object exactly as for the ePIE:

$$\begin{aligned} P'(\mathbf{r} - \mathbf{R}_c) &= \mathcal{U}[P(\mathbf{r} - \mathbf{R}_c), O_1(\mathbf{r}), \Delta\psi(\mathbf{r})], \\ O'_1(\mathbf{r}) &= \mathcal{U}[O_1(\mathbf{r}), P(\mathbf{r} - \mathbf{R}_c), \Delta\psi(\mathbf{r})]. \end{aligned}$$

14. Set $O_n(\mathbf{r}) = O'_n(\mathbf{r})$ for every n and set $P(\mathbf{r}) = P'(\mathbf{r})$.

Steps 1–14 repeat until all of the recorded diffraction patterns have been addressed, which completes a single iteration of the algorithm. Further iterations follow, until either a terminating condition is met or a fixed number of iterations are completed.

C. Uniqueness

In common with 2D ptychography, the 3PIE attempts to satisfy the following three constraints that encapsulate *a priori* knowledge of the ptychographic experiment: the specimen does not change during data collection, the incident probe does not change during data collection, and the reconstructed probe and object slices, when inputted to the forward propagation model of Fig. 1(a), produce at each probe position a wavefront at the detector plane whose intensity matches the corresponding measured diffraction pattern. If the 3PIE reaches a set of reconstructed object slices and a reconstructed probe that together give exactly the measured diffraction patterns at each probe offset, the algorithm will not then deviate from this solution.

However there will in general be more than one solution for the recovered object layers that satisfy these constraints. For instance, if the correct reconstruction of the N object slices is $O_n(\mathbf{r})$ (again with $n = 1 \dots N$) then the reconstruction $c_n O_n(\mathbf{r})$, where each c_n is a complex constant and $\prod_{n=1}^N c_n = 1$, is also a valid solution, meaning that the

complex transmittance of the reconstructed slices can be related one to another only up to a constant scale factor. This is identical to the case of 2D ptychography, where the scaling of the intensity of the reconstructed image is arbitrary and the reconstructed phase is relative, not absolute. An additional constraint restricting the maximum modulus of each slice to less than unity can be employed to mitigate this ambiguity; furthermore, as is the case in two-dimensions, if an area of free-space exists within the field of view of the reconstruction, such as at the edge of the specimen, a modulus and phase can be associated with this region that links together the scaling of the object slices, destroying the above ambiguity and providing an absolute measure of both intensity and phase relative to free-space.

A second ambiguity arises when two adjacent slices of the object, say $O_n(\mathbf{r})$ and $O_{n+1}(\mathbf{r})$, are in close proximity, well within the bounds of the multiplicative approximation. Then $(f(\mathbf{r}) \cdot O_n(\mathbf{r}), f^{-1}(\mathbf{r}) \cdot O_{n+1}(\mathbf{r}))$, where $f(\mathbf{r})$ is any complex-valued function, will also be a solution. When this is the case the two slices should clearly be combined into a single slice; therefore the slice separations must be carefully chosen given the experimental setup from which data is collected.

3. VISIBLE LIGHT EXPERIMENTAL RESULTS

We demonstrate 3PIE experimentally using visible light—Fig. 2 shows our experimental arrangement. Reconstructions were carried out using a MATLAB implementation of the 3PIE employing GPU-based acceleration. The 100 iterations of ePIE carried out to produce Figs. 3(a)–3(b) took 140 s, while the 3PIE reconstructions of Figs. 3(c)–3(f) took 232 s. The reconstruction time increased roughly linearly with the number of slices and iterations thereafter.

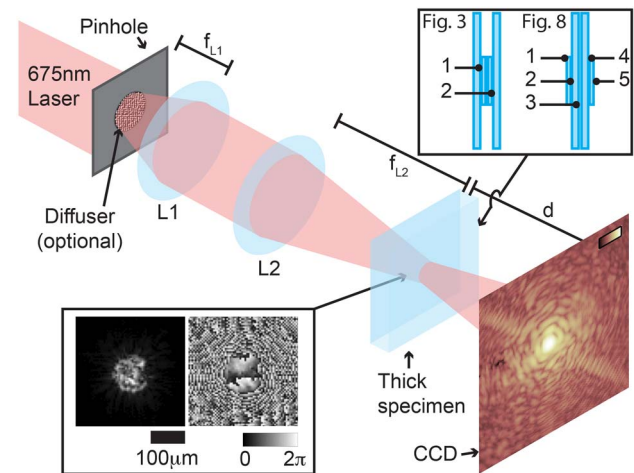


Fig. 2. (Color online) Experimental setup for collection of ptychographic data. The optional diffuser consisted of a thin piece of plastic film covering the $100\ \mu\text{m}$ mounted pinhole. Lens L1 was an antireflection coated doublet with $f_{L1} = 3\ \text{cm}$. Lens L2 was a matching doublet during the experiments resulting in Figures 3, 5 and 8, but was replaced by a $10\times$ Olympus microscope objective ($f_{L2} = 18\ \text{mm}$) for the experiment resulting in Figs. 6 and 7. The distance d was accurately measured before each experiment. A typical diffraction pattern from the system is shown to the right on a logarithmic scale. The lower inset shows the modulus and phase of a typical probe function reconstructed by the 3PIE. The upper inset shows the arrangement of the two microscope slides used as specimens for Fig. 3 (left) and Fig. 8 (right), together with the planes used for the 3PIE reconstructions.

A. Details of Experimental Procedure

All of our experiments were carried out on the same experimental rig. A 675 nm fiber-coupled laser diode module supplied the illumination. When present, the (static) diffuser consisted of a thin sheet of plastic film attached to the 100 μm pinhole. Its purpose was to reduce the dynamic range of the signal incident upon the detector by transforming the intense zero-order undiffracted beam (which usually results in an intense spot at the centre of each diffraction pattern) into a speckle pattern distributed over a large proportion of the detector. This makes recording of the diffraction patterns easier, quicker, and more accurate [27]. Two standard one-inch diameter doublet lenses, both with a 30 mm focal length, were used for the results presented in all but Figs. 6 and 7; for these figures the downstream doublet (L2) was replaced with a 10 $\times$ Olympus objective lens. In the first configuration the two doublets were arranged in a 4F configuration to image the 100 μm pinhole onto the specimen with no magnification. When L2 was replaced by the objective lens, a 0.6 $\times$ demagnified image of the pinhole (around 60 μm in diameter) was incident upon the sample. This allowed a smaller separation between the slices of the reconstruction. In x ray and electron microscopy the probe would be formed using a lens to focus a beam to a small spot rather than to image a pinhole, but in the

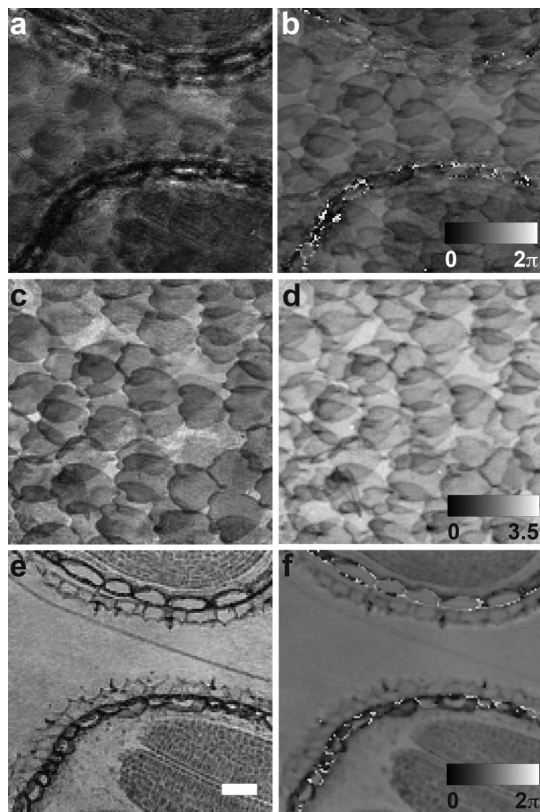


Fig. 3. Final reconstructions of a butterfly wing and capsella embryo. The progress of this reconstruction is shown in (Media 1). The specimens were placed coverslip to coverslip and ptychographic data collected. The conventional ePIE reconstruction yielded the modulus and phase shown in (a) and (b). The 3PIE reconstruction was carried out at two planes coincident with the two specimens (see upper inset of Fig. 2) and produced the moduli and phases of (c) and (d) at the first plane and (e) and (f) at the second plane. Scale bar is 100 μm .

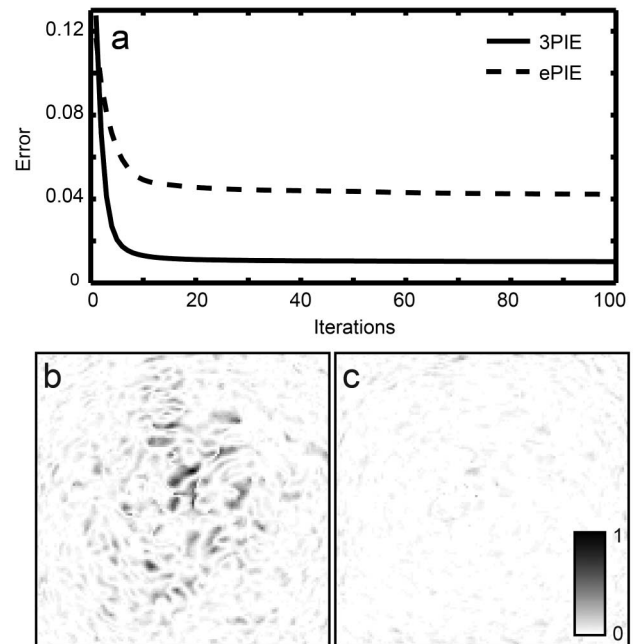


Fig. 4. Improvement realized by moving from a projection-based propagation model to a multi-slice model; (a) The evolution of the error metric given by Eq. (4) of the main text as the ePIE and 3PIE reconstructions shown in Fig. 3 progress; (b) The error between one of the measured diffraction patterns and the corresponding pattern as generated by the ePIE reconstruction and its multiplicative model; (c) The same error measure calculated using the 3PIE reconstruction and its multi-slice model. Parts (b) and (c) are plotted on identical grey scales.

visible light regime this would remove our ability to use the diffuser.

The specimen, rather than the probe, was translated in our experiments. To this end, specimens were mounted on a PI M-686 two-axis piezo stage, which was programmed to translate the specimens in a grid with a nominal pitch of 20 μm (6 μm for Figs. 6 and 7), with each position offset by a random component between $\pm 5\%$. A 14 bit AVT Pike CCD camera, with 2048×2048 pixels on a 7.4 μm pitch, was positioned approximately $d = 70$ mm ($d = 32$ mm for Figs. 6 and 7) downstream of the specimen. This distance was measured accurately before each set of data was collected.

Although the diffuser considerably reduced the dynamic range of the diffraction patterns, by taking four exposures of different lengths at each specimen position, noise was further reduced. The exposures were appropriately scaled and combined to form a single diffraction pattern for each specimen position, which were then binned down to 128×128 pixels. This arrangement gave a pixel pitch in the reconstructed images of around 3 μm (1.4 μm for Figures 6 and 7).

B. Experimental Results

For the images shown in Fig. 3 our specimen consisted of two microscope slides arranged coverslip to coverslip, as shown in the upper inset of Fig. 2. Mounted on the first slide was a section from a butterfly's wing and on the second a capsella embryo. The measured separation between the butterfly wing and the capsella was 380 μm , corresponding to the thickness of the two coverslips, and the sample-detector distance was $d = 69$ mm.

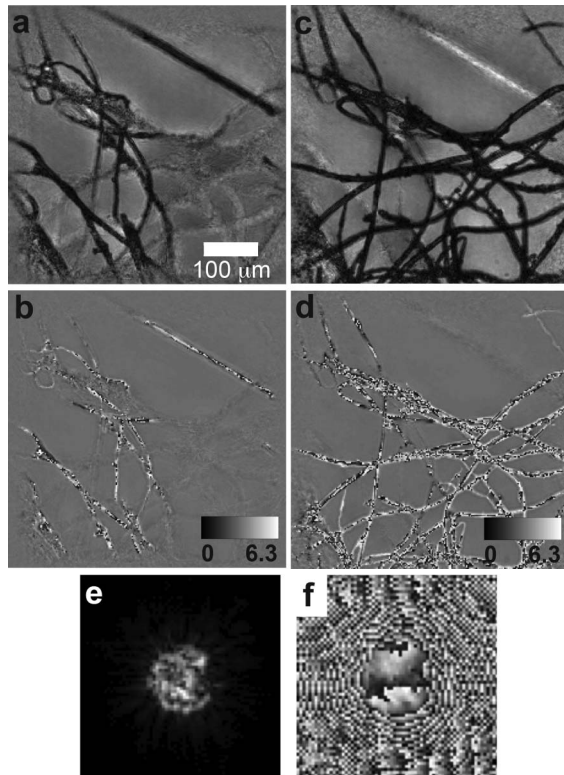


Fig. 5. Ptychographic data were collected from a piece of tissue paper in this experiment. 3PIE separated the fibers of the tissue at two planes; (a) and (b) modulus and phase of the downstream plane, (c) and (d) modulus and phase of the upstream plane. (e) and (f) the reconstructed probe.

A dataset was collected from $N = 400$ specimen positions arranged on a 20×20 grid, and initially reconstructed using the ePIE; this produced the modulus and phase images shown in Figs. 3(a) and 3(b). That the reconstruction has not failed entirely indicates that the multiplicative approximation of Eq. (1) was “almost” satisfied in this case; however the result is a blurred representation of both the wing and the capsella, with their finer features lost. For the 3PIE reconstruction the same data were used, but the reconstruction was split into two slices, separated by $380 \mu\text{m}$ so that each slice was coincident with one of the mounted specimens. 100 iterations of the 3PIE were carried out with the α constant of the update function [Eq. (3)] set to 1 in every instance. To correctly model the propagation of the wavefront between the slices of the reconstruction, in this and subsequent experiments the wavelength inputted to the 3PIE was adjusted to account for the refractive index of the glass slides ($n = 1.518$) upon which the specimens were mounted. Figure 3 (Media 1) shows the evolution of the two image slices as the reconstruction progresses, demonstrating that good separation of the two images is realized after only 20 or so 3PIE iterations. Figures 3(c)–3(f) show the final 3PIE results: at the first slice clean modulus and phase images [Figs. 3(c)–3(d)] of the butterfly wing are reconstructed, and clear modulus and phase images [Figs. 3(e)–3(f)] of the capsella result at the second. The algorithm also reconstructs the probe wavefront incident upon the first slice; its modulus and phase are shown in the lower inset to Fig. 2.

Figure 4 illustrates how, for the butterfly/capsella two-layer specimen, the 3PIE and its multi-slice propagation model

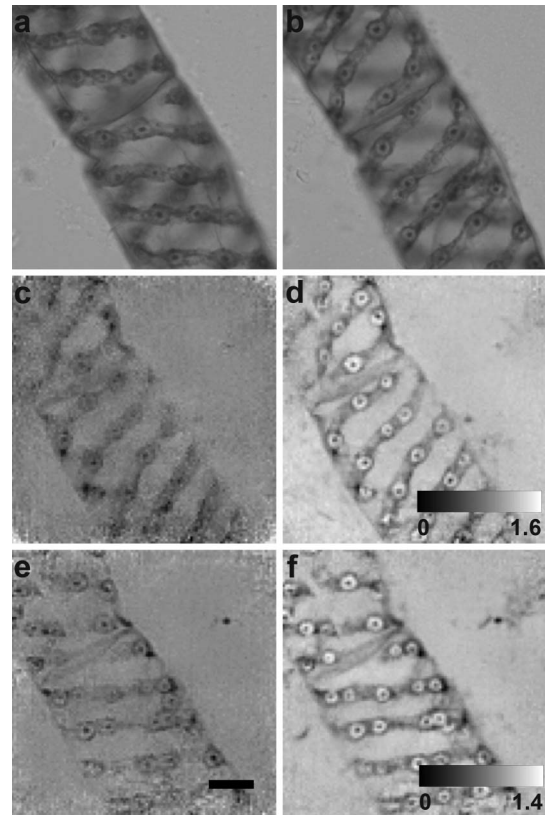


Fig. 6. Reconstruction of a spirogyra. A higher illumination NA (formed using a $10\times$ microscope objective) and a shorter camera length ($d = 32 \text{ mm}$) were used in the setup from which these reconstructions were generated. (a) and (b) white light brightfield images focusing on (a) the top and (b) the bottom of the spirogyra; (c) modulus and (d) phase of the first slice of the 3PIE reconstruction, focused at the bottom of the spirogyra; (e) modulus and (f) phase of the second slice, focused at the top. Scale bar is $25 \mu\text{m}$.

improves over the ePIE and its multiplicative approximation model. In Fig. 4(a) the diffraction patterns produced after k iterations of the two algorithms are compared to the measured diffraction data, according to the usual error metric [12]:

$$E(k) = \frac{\sum_{j=1}^J \sum_{\mathbf{u}} \left(|\Psi_{j,k}(\mathbf{u})| - \sqrt{I_j(\mathbf{u})} \right)^2}{\sum_{\mathbf{u}} I_j(\mathbf{u})}, \quad (4)$$

where $\Psi_{j,k}(\mathbf{u})$ denotes the wavefront $\Psi_j(\mathbf{u})$ after k iterations of either algorithm. Figures 4(b) and 4(c) compare a single measured diffraction pattern (the 210th, at the center of the reconstruction) to the corresponding patterns generated from (b) the ePIE and (c) the 3PIE reconstructions of Fig. 3, according to the metric $E(\mathbf{u}) = \sum_{\mathbf{u}} (|\Psi_{210}(\mathbf{u})| - \sqrt{I_{210}(\mathbf{u})})^2$ (the plots are normalized to the same colour scale). It is evident from these results that by accounting for 3D and multiple scattering effects, the multi-slice approach has provided a more precise model of the way light propagated in the actual experiment. It is also clear that the multiplicative approximation is not a sufficiently accurate model in the case of this two-layered specimen, that although the ePIE did not fail completely, the resulting reconstruction is an artifice from which it is not possible to extract accurate images of either the butterfly wing or the capsella. This point should be emphasized: if quantitative

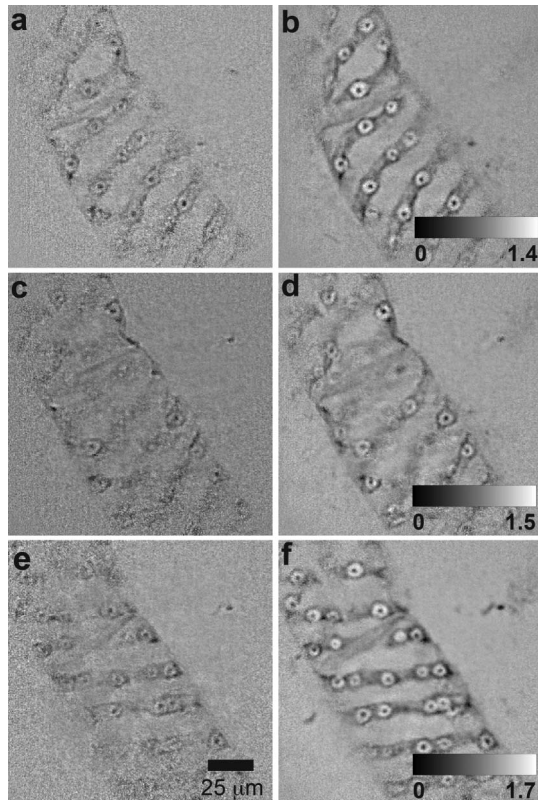


Fig. 7. Second three-plane reconstruction from the spirogyra data used in Fig. 6. Here the 3PIE was combined with a ptychographic super-resolution method to both increase the resolution of the reconstruction and to allow a third slice, between the previous two, to be introduced. The spacing between the slices in this reconstruction is $11\ \mu\text{m}$.

images are to be reconstructed, the effect of the butterfly wing on the wavefront incident upon the capsella *must* be accounted for—it is not possible to carry out two independent ptychographic reconstructions simply by propagating a probe wavefront to the planes of the two specimens.

To further demonstrate the operation of the 3PIE a second scan from the same experimental setup as that used to collect the data for Fig. 3 was undertaken. In this instance the two microscope slides were replaced by a piece of tissue and the reconstruction carried out using a slice spacing of $100\ \mu\text{m}$. Figures 5(a)–5(b) show the modulus and phase of the downstream slice and Figs. 5(c)–5(d) those of the upstream slice. The recovered probe is shown in Figs. 5(e)–5(f). The algorithm has identified which of the various tissue fibers lie within each slice volume, with those that cross between the slices blurring from one as they pass into the next. Despite the almost total attenuation by the tissue fibers of the incident illumination, a reasonable recovery of free-space in the fiber's 'shadow' has been achieved.

In our third example, lens L2 (see Fig. 2) was replaced by a microscope objective to realize a larger NA in the illumination optics, and the sample-detector distance was reduced to $d = 32\ \text{mm}$. This resulted in a smaller probe and higher image resolution that allowed us to reduce the distance between the slices in our reconstruction. The specimen used was a spirogyra filament (a type of algae): this is normally cylindrical in shape with a characteristic helix around its surface, but in these experiments the specimen was compressed

by the cover slip so its sides were flattened and were separated by $22\ \mu\text{m}$. Figures 6(a) and 6(b) show images of the spirogyra as recorded by a brightfield microscope fitted with the same objective lens that was utilized in the 3PIE experimental rig—for Fig. 6(a) the microscope focus was adjusted to the bottom surface of the spirogyra and in Fig. 6(b) to the top. Clearly the depth of field of the objective is too large to section the two sides of the ribbon-like chloroplasts that form the compressed helix. In contrast, Figs. 6(c) and 6(f) show successful sectioning of the top and bottom of the algae's chloroplast skeleton after 100 iterations of the 3PIE (using 2 slices separated by $22\ \mu\text{m}$ and with $\alpha = 1$ for every instance of the update function). Figure 7 shows an additional reconstruction carried out using the same data as that used to generate Fig. 6, incorporating a third slice halfway between the two slices shown in that figure. For this reconstruction the diffraction patterns were extrapolated to $2.5\times$ the extent of the recorded data using a super-resolution method [28]. This allowed us to further reduce the minimum separation of the slices in our reconstruction to $11\ \mu\text{m}$, and extract the sides of the spirogyra helix in the additional slice.

A fourth experiment extended to five the number of slices in the 3PIE reconstruction. Microscope slides containing the butterfly's wing and an insect's mouth parts were arranged back to back, so that the two specimens were around $2\ \text{mm}$ apart: the arrangement of the slides and the planes of the reconstructed object slices are shown in the upper inset of Fig. 2. The diffuser was removed for this experiment as the specimen itself scattered so strongly that the dynamic range of the recorded diffraction patterns was acceptable without it. It was also difficult with the diffuser present to sufficiently localise the probe over the large z -range of the specimen used in this experiment. The sample-detector distance here was measured as $d = 67.8\ \text{mm}$, and ptychographic data was collected from 20×20 specimen positions.

The conventional 2D ePIE did not converge to a sensible solution using the data from this experiment, indicating a total breakdown of the multiplicative approximation. In contrast, Fig. 8 shows the moduli and phases of the five planes of the 3PIE reconstruction. The butterfly wing and insect (predominantly its eye) are well resolved in Figs. 8(c) and 8(d) and 8(g) and 8(h). The reconstructions coinciding with the coverslips show numerous specks of dust [Figures 8(a) and 8(b) and 8(i)–8(j)], and a small amount of crosstalk due to the proximity of these slices to their neighbours. The touching faces of the microscope slides did not hold many features; however the algorithm still faithfully recovers predominantly empty space, together with a few small dust specks and scratches [Fig. 8(e)–8(f)].

The small separation of the outermost two slices meant that clean images at these planes were difficult to realize. A good strategy for the reconstruction in this case was to perform an initial number of iterations with the object update function constant $\alpha = 0$ in the update of these outer slices, and $\alpha = 1$ for the central three slices. 100 iterations of 3PIE were carried out under these conditions, before α was changed to 1 for the update of both outer slices and a further 250 iterations performed. Without this addition the 3PIE was not able to remove a blurry shadow of the insect proboscis from the final slice of the reconstruction, which evidently lies very close to the limit of slice separation for this experimental geometry.

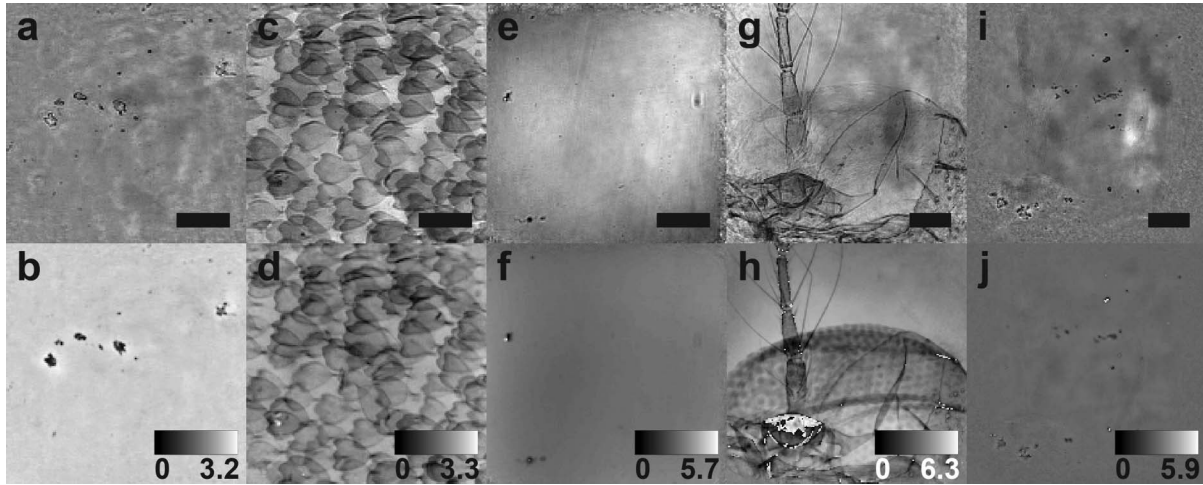


Fig. 8. 3PIE reconstruction using five object planes. In this example two slides were placed back to back and the reconstruction carried out over the five planes labelled in the upper inset of Fig. 2. The progress of a similar reconstruction is shown in (Media 2); (a) and (b) modulus and phase of the first slice, coincident with dust on a coverslip; (c) and (d) modulus and phase of the second slice, 180 μm downstream and coincident with a butterfly wing sample; (e) and (f) third slice, 1.1 mm downstream, coincident with the two touching faces of the microscope slides; (g) and (h) fourth slice, 1.2 mm downstream, coincident with an insect sample, showing mainly its eye. (i) and (j) fifth slice, 160 μm downstream, coincident with the second coverslip. The scale bars are 100 μm .

The proximity of the slices also resulted in slightly brighter patches appearing in the featureless regions of the final slice, with corresponding darker patches appearing in the preceding slice. This problem has not been observed except when the slices are in close proximity.

Figure 8 (Media 2) shows an example reconstruction from the data used to generate Fig. 8. As described above the outermost two slices are not updated for the first 100 iterations and the specks of dust therefore appear in the slices containing the butterfly wing and insect eye. After 100 iterations the first slice starts to update and the specks of dust contaminating the butterfly wing image are slowly drawn into the correct slice. After 150 iterations, update of the last slice begins and the specks of dust seen in the insect eye image are likewise extracted.

Figure 9 shows how the modulus of the probe is affected as it passes through each slice of the reconstruction shown in

Fig. 8. The figure demonstrates that there is significant variation introduced into the probe during its passage through the specimen and that it cannot be considered under a single-scattering approximation.

4. DISCUSSION

The benefits of the 3PIE can be highlighted by making an important distinction between ptychography and alternative coherent imaging techniques such as digital holography and single-shot coherent diffractive imaging. The aim of these methods is to reconstruct a complex-valued wavefront based upon a recording of its intensity at the plane of a detector. This wavefront can then be propagated arbitrarily to a plane upstream, within or downstream of the specimen. When the specimen is 3D, varying the z plane of the wavefront is equivalent to racking through defocus on a conventional microscope: out

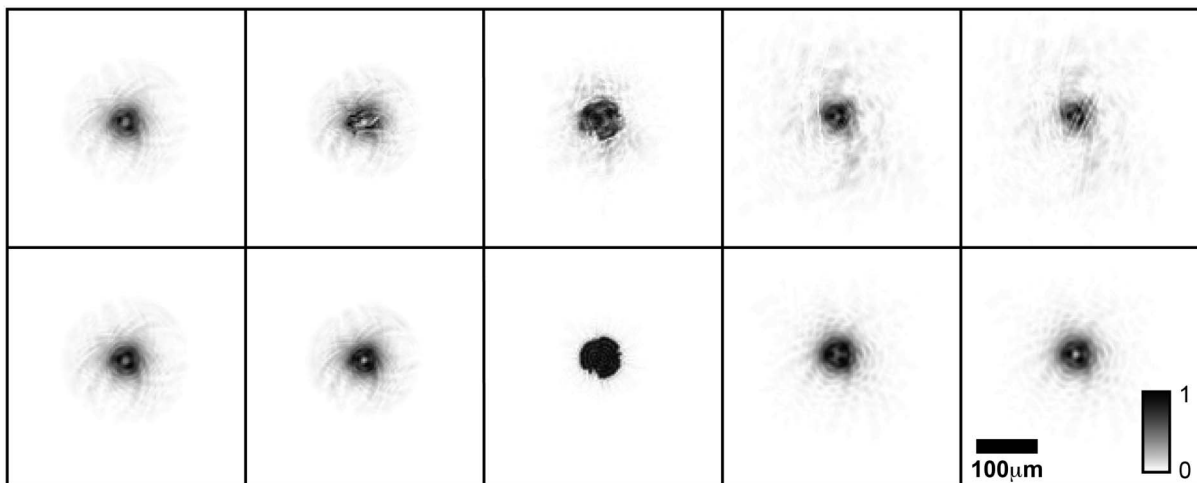


Fig. 9. Evolution of the exit wave, $\psi_{e,n}(\mathbf{r})$, as the forward calculation of the 3PIE passes through the five slices shown in Fig. 8. Top row: the modulus of $\psi_{e,n}(\mathbf{r})$ for (left) $n = 1$ through to (right) $n = 5$. Bottom row: the reconstructed probe wavefront propagated to the same five reconstruction planes without the specimen present. The comparison shows the significant effect of the specimen on the illuminating probe as it propagates through the layers of the two microscope slides.

of focus planes of the object confuse the image, especially when strong multiple scattering is present. This is further exacerbated for coherent illumination where out of focus features cause speckle, a problem that short coherence length digital holography [23] attempts to remedy. In contrast, ptychography does not reconstruct a wavefront. A ptychographic data set consists of intensity measurements of a multitude of wavefronts, and a ptychographic reconstruction is a complex-valued transmission function that is consistent with all of these wavefronts. It is only in the case of a 2D specimen (that is, one to which the multiplicative approximation can be applied) that the ptychographic reconstruction is equivalent to the wavefront that would have resulted were the specimen illuminated by a plane wave. A 3PIE reconstruction is not a representation of the exit wave from a 3D specimen; it is a set of transmission functions that, when combined with the multi-slice propagation model, satisfy the constraints set out in Subsection 2.C.

The most naïve approach to understand the source of 3D information upon which the 3PIE draws is to consider the ray diagram illustrated in Fig. 10. For convergent or divergent illumination a shadow image is projected onto the far-field detector plane. We see that in general no single 2D object can be consistent with a set of shadow images collected from overlapping regions of the 3D specimen as the probe is shifted: this is because features at different depths within the object will move across the detector plane at different rates as a function of probe shift (for a 2D object, all features move by an equal distance for a given probe shift). The progress of the shadow of any particular element within the object can be used to determine its depth.

In the wave formulation similar arguments apply. One might suppose that for a 3D specimen it would be possible to undertake a conventional ptychographic reconstruction of a particular layer within it using a probe function calculated from its free-space propagation at that layer and then build up a 3D representation from independent reconstructions of

many such layers. However, any one of the measured diffraction patterns is composed of a convolution of the many wavefronts scattered from all layers of the specimen, meaning that the intensity of each diffraction pattern bears no relation to what would have been measured from any one layer. In other words, as in the ray optic approach, the detector plane expresses information from all layers of the specimen simultaneously: only by taking into account the 3D and multiple scattering effects can a 3D ptychographic reconstruction ever be made consistent with the measured data.

5. CONCLUSION

The key benefit of this new form of 3D imaging is its application to thick specimens, where multiple scattering prevents conventional methods from being applied; it is immediately applicable to imaging the front and back surfaces of bulk material specimens, for example, imaging patterned semiconductor wafers using infrared light. A further benefit is the relatively high contrast provided by the phase image. This will be important for thick biological samples where absorption within any one voxel of the object is small and where, unlike confocal microscopy, the phase and absorption sensitivity can be used to infer the complex dielectric constant of each voxel of the object. The experimental setup is simple and removes the requirement to rotate the sample (as in computed tomography) or scan it along the optical axis (as in confocal microscopy). It may also be possible to combine our approach with specimen rotations to image very strongly scattering specimens at high resolution in three dimensions where conventional back-projection methods would fail.

Much of the depth resolution in our technique derives from the range of incident beam angles within the probe, each of which explores a different projection of the specimen. However, the amount of data collected for a given field of view (principally a function of how far the probe is stepped for each probe position) also impacts upon how much 3D information can be practically extracted from the size of the recorded data set [28]. We have not yet comprehensively tested the limits on the depth resolution achievable, the number of slices that can be reconstructed or the thickness of specimen that can be accommodated. All this will be the subject of future work.

Implementing 3PIE at the smaller length scales of x ray and electron microscopy will be considerably more difficult than the visible-light experiments we have so far carried out; however we envisage that our technique will have wide applications in many fields of microscopic structural discovery and will be especially relevant to electron microscopy, where multiple scattering effects are endemic and seriously impair atomic-scale imaging.

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REFERENCES

1. W. Hoppe, “Diffraction in inhomogeneous primary wave fields: 1. Principle of phase determination from electron diffraction interference,” *Acta Crystallogr. Sec. A* **25**, 495–501 (1969).

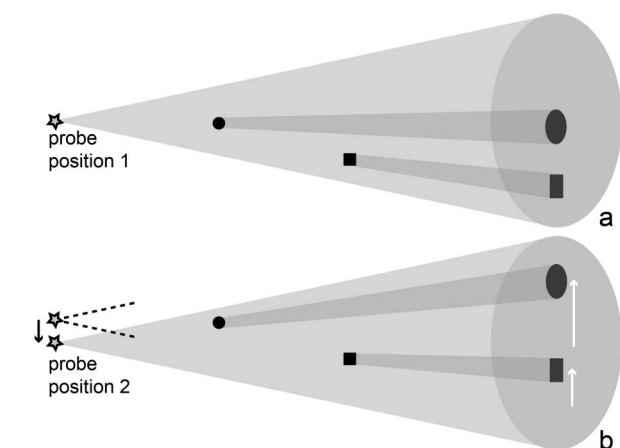


Fig. 10. Simple ray-optic schematic illustrating how 3D information is encoded in a ptychographic scan. The circle and square represent the 3D specimen. At a first probe position (a) the detector records a projection of the circle and square—without diffraction effects there is no 3D information encoded in this recording. However when the probe is moved to a second position (b), and a second recording made, parallax has caused the projections of the circle and square to move by different amounts—this relative movement can be used to deduce their position along the optical axis.

2. K. Giewekemeyer, P. Thibault, S. Kalbfleisch, A. Beerlink, C. Kewish, M. Dierolf, F. Pfeiffer, and T. Salditt, "Quantitative biological imaging by ptychographic x-ray diffraction microscopy," *Proc. Natl. Acad. Sci. U.S.A.* **107**, 529–534 (2010).
3. P. Thibault and V. Elser, "X-ray diffraction microscopy," *Annual Review Condensed Matter Physics* **1**, 237–255 (2010).
4. A. Schropp, P. Boye, A. Goldschmidt, S. Hönig, R. Hoppe, J. Patommel, C. Rakete, D. Samberg, S. Stephan, S. Schöder, M. Burghammer, and C. G. Schroer, "Non-destructive and quantitative imaging of a nano-structured microchip by ptychographic hard X-ray scanning microscopy," *J. Microsc.* **241**, 9–12 (2011).
5. K. Giewekemeyer, M. Beckers, T. Gorniak, M. Grunze, T. Salditt, and A. Rosenhahn, "Ptychographic coherent x-ray diffractive imaging in the water window," *Opt. Express* **19**, 1037–1050 (2011).
6. S. Hönig, R. Hoppe, J. Patommel, A. Schropp, S. Stephan, S. Schöder, M. Burghammer, and C. G. Schroer, "Full optical characterization of coherent x-ray nanobeam by ptychographic imaging," *Opt. Express* **19**, 16324–16329 (2011).
7. F. Hübner, J. M. Rodenburg, A. M. Maiden, F. Sweeney, and P. A. Midgley, "Wave-front phase retrieval in transmission electron microscopy via ptychography," *Phys. Rev. B* **82**, 121415 (2010).
8. C. T. Putkunz, A. J. D'Alfonso, A. J. Morgan, M. Weyland, C. Dwyer, L. Bourgeois, J. Etheridge, A. Roberts, R. E. Scholten, K. A. Nugent, and L. J. Allen, "Atom-scale ptychographic electron diffractive imaging of boron nitride cones," *Phys. Rev. Lett.* (in press).
9. M. J. Humphry, B. Kraus, A. C. Hurst, A. M. Maiden, and J. M. Rodenburg, "Ptychographic electron microscopy using high-angle dark-field scattering for sub-nanometre resolution imaging," *Nat. Commun.* **3**, 730 (2012).
10. M. Guizar-Sicairos and J. R. Fienup, "Phase retrieval with transverse translation diversity: a nonlinear optimization approach," *Opt. Express* **16**, 7264–7278 (2008).
11. P. Thibault, M. Dierolf, A. Menzel, O. Bunk, C. David, and F. Pfeiffer, "High-resolution scanning x-ray diffraction microscopy," *Science* **321**, 379–382 (2008).
12. A. M. Maiden and J. M. Rodenburg, "An improved ptychographical phase retrieval algorithm for diffractive imaging," *Ultramicroscopy* **109**, 1256–1262 (2009).
13. H. N. Chapman, A. Barty, S. Marchesini, A. Noy, S. P. Hau-Riege, C. Cui, M. R. Howells, R. Rosen, H. He, J. C. Spence, U. Weierstall, T. Beetz, C. Jacobsen, and D. Shapiro, "High-resolution ab initio three-dimensional x-ray diffraction microscopy," *J. Opt. Soc. Am. A* **23**, 1179–1200 (2006).
14. M. Dierolf, A. Menzel, P. Thibault, P. Schneider, C. M. Kewish, R. Wepf, O. Bunk, and F. Pfeiffer, "Ptychographic x-ray computed tomography at the nanoscale," *Nature* **467**, 436–439 (2010).
15. M. Guizar-Sicairos, A. Diaz, M. Holler, M. S. Lucas, A. Menzel, R. Wepf, and O. Bunk, "Phase tomography from x-ray coherent diffractive imaging projections," *Opt. Express* **19**, 21345–21357 (2011).
16. T. Plamann and J. M. Rodenburg, "Double resolution imaging with infinite depth of focus in single lens scanning microscopy," *Optik* **96**, 31–36 (1994).
17. J. M. Rodenburg, "Can Ronchigrams provide a route to sub-angstrom tomographic reconstruction?" *Inst. Phys. Conf. Ser.* **179**, 185–190 (2004).
18. P. Godard, G. Carbone, M. Allain, F. Mastropietro, G. Chen, L. Capello, A. Diaz, T. H. Metzger, J. Stangl, and V. Chamard, "Three-dimensional high-resolution quantitative microscopy of extended crystals," *Nat. Commun.* **2**, 568 (2011).
19. R. McIntosh, D. Nicastro, and D. Mastronarde, "New views of cells in 3D: an introduction to electron tomography," *Trends Cell Biol.* **15**, 43–51 (2005).
20. S. Popoff, G. Leroose, M. Fink, A. C. Boccara, and S. Gigan, "Image transmission through an opaque material," *Nat. Commun.* **1**, 81 (2010).
21. J. M. Schmitt, A. Knüttel, and M. Yadlowsky, "Confocal microscopy in turbid media," *J. Opt. Soc. Am. A* **11**, 2226–2235 (1994).
22. J. Garcia-Sucerquia, W. Xu, S. K. Jericho, P. Klages, M. H. Jericho, and H. Jürgen Kreuzer, "Digital in-line holographic microscopy," *Appl. Opt.* **45**, 836–850 (2006).
23. L. Martínez-León, G. Pedrini, and W. Osten, "Applications of short-coherence digital holography in microscopy," *Appl. Opt.* **44**, 3977–3984 (2005).
24. J. M. Rodenburg, A. M. Maiden, and M. J. Humphry, "Improvements in three dimensional imaging," *International Patent Application Number WO2012/038749* (2011).
25. J. M. Cowley and A. F. Moodie, "The scattering of electrons by atoms and crystals. 1. a new theoretical approach," *Acta Crystallogr. Sec. A* **10**, 609–619 (1957).
26. E. J. Kirkland, *Advanced Computing in Electron Microscopy* 2nd ed. (Springer, 2010), Chap. 6.
27. A. M. Maiden, J. M. Rodenburg, and M. J. Humphry, "Optical ptychography: A practical implementation with useful resolution," *Opt. Lett.* **35**, 2585–2587 (2010).
28. A. M. Maiden, M. J. Humphry, F. Zhang, and J. M. Rodenburg, "Superresolution imaging via ptychography," *J. Opt. Soc. Am. A* **28**, 604–612 (2011).